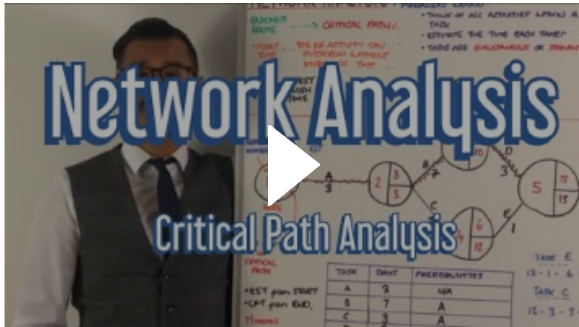


4 - Critical Path Analysis

Flipped learning - Please watch the short video below before the start of the lesson



Learning Outcome

To be able to complete and interpret critical path analysis diagrams and assess their value in project planning

To Know



- The purpose and benefits of using Critical Path Analysis in project management.
- How to complete a simple network diagram and identify the critical path.
- How to calculate:
 - Earliest Start Times (EST)
 - Latest Finish Times (LFT)
 - Total Float
- How to interpret the results to plan and prioritise activities effectively.
- Discuss the limitations of CPA (e.g. reliance on accurate estimates, doesn't consider resource constraints).

Critical path analysis is a technique used to identify the **order in which all activities need to be completed** when planning a complex project.

Critical path diagrams organise the activities in an order to show which activities can be **done simultaneously and which are dependent upon earlier activities.**

This allows for the identification of the **shortest time in which a project can be completed.**

Building the CPA Diagram

The technique for CPA requires a business to construct a model of the project that includes the following:

A list of all activities required to complete the project

The time (duration) that each activity will take to complete

The dependencies between the activities (i.e. one activity must be finished before the next activity can begin)



I Do, We Do, You Do

Task	Activity	Order	Duration (months)
A	Conduct customer research	Starting activity	2
B	Design product concept	Begin when A complete	4



06/22

Extract F

Coca-Cola is giving one of its most popular drinks a makeover

Coca-Cola is changing the recipe for its Coca-Cola Zero Sugar in an effort to make the drink taste more like regular Coke. The Coca-Cola Zero Sugar aluminium can will also look different – all red rather than the present red and black can.

The move comes as Coca-Cola focuses its resources on its trademark products, including Coke, Coca-Cola Zero Sugar and Diet Coke after stopping production of some struggling brands. The company hopes the updated version of Coca-Cola Zero Sugar will attract more consumers.

The new version of Coca-Cola Zero Sugar is estimated to replace the current one in the US and Canada in September 2021.

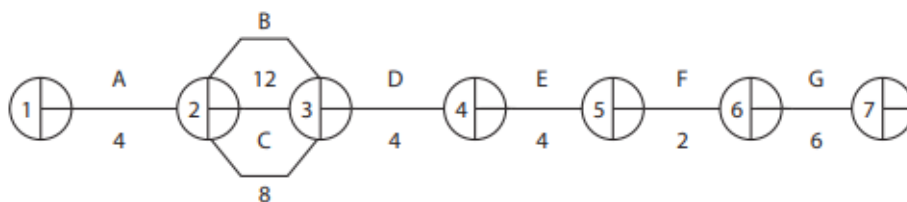
5

10

Extract G

Proposed schedule and network for the redevelopment of Coca-Cola Zero Sugar

	Activity	Duration (weeks)	Preceding Activities
A	Conduct market research	4	-
B	Develop new recipe	12	A
C	Design new drinks can	8	A
D	Test marketing	4	B, C
E	Develop final recipe	4	D
F	Distribute final recipe	2	E
G	Promotional activity	6	F



Using the data in Extracts F and G, and appropriate critical path analysis (CPA) calculations, evaluate the benefits of CPA to a business such as Coca-Cola.

(20)



Affirmative Checking

Paragraph 1

POINT

One advantage is....

IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE STATED BUSINESS OBJECTIVE?	This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 2 POINT IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Another advantage is ... This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 3 POINT IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE STATED BUSINESS OBJECTIVE?	However, one disadvantage/limitation/point to consider is.... This is because... Consequently/This means/This will lead to... Therefore... As a result, ...
Paragraph 4 POINT	Another disadvantage/limitation/point to consider is.... This is because...

IMPLICATION/S WHAT IMPACT DOES THE IMPLICATION/S HAVE ON THE BUSINESS OBJECTIVE/S?	Consequently/This means/This will lead to... Therefore... As a result, ...
Conclusion: <u>4W's</u> Which option? Why? What factors does the success of your chosen option depend upon?	Overall, (directly answer the question asked) This is because... Success will depend on... It is recommended that...

What are the advantages and disadvantages of using CPA as a decision making tool?



